

gives the *relative frequency* house owners and can be used as an estimate of the true P_i corresponding to each X_i . This true P_i is used to estimate the logit, $\hat{L}_t$.

$$\text{Thus, } \hat{L}_i = \ln\left(\frac{\hat{p}_i}{1-\hat{p}_i}\right) = \hat{\alpha}_0 + \hat{\alpha}_1 X_i \quad \dots (15.14)$$

gives a fairly good estimate of the true logit, L_i , if the number of observations N_i at each X_i is reasonably large.

Notes on the estimation of Logit model:

- Logit models are estimated using maximum likelihood method (MLM).
- Standard errors (SE) are asymptotic (i.e., applicable for large samples). Hence, we have to use Z statistic instead of t-statistic.
- R-squared is not meaningful in logit models.
- Likelihood ratio (LR) test is applicable.

15.3.2 Interpretation of Logit Model

(Example 15.2)

Table 15.3: Cross-sectional Data on GPA, ASE, PC and GRADE

Students	GPA	ASE	PC	GRADE (A=1, rest=0)
1	2.66	20	0	0
2	2.89	22	0	0
3	3.28	24	0	0
4	2.92	12	0	0
5	4	21	0	1
6	2.86	17	0	0
7	2.76	17	0	0
8	2.87	21	0	0
9	3.03	25	0	0
10	3.92	29	0	1
11	2.63	20	0	0
12	3.32	23	0	0
13	3.57	23	0	0
14	3.26	25	0	1
15	3.53	26	0	0
16	2.74	19	0	0
17	2.75	25	0	0
18	2.83	19	0	0
19	3.12	23	1	0
20	3.16	25	1	1
21	2.06	22	1	0
22	3.62	28	1	1
23	2.89	14	1	0
24	3.51	26	1	0
25	3.54	24	1	1
26	2.83	27	1	1
27	3.39	17	1	1
28	2.67	24	1	0
29	3.65	21	1	1
30	4	23	1	1
31	3.1	21	1	0
32	2.39	19	1	1

Suppose that a Mathematical Economics instructor gave routine weekly exercises to her section “A” students in the previous semester. She wants to study the effect of the performance in these routine exercises on the final grade of students in the

course, Mathematical Economics. At the end of the semester, she found average scores in exercise (ASE) for each student.

The dependent variable is the final grade of students for Mathematical Economics ($Y_i = 1$, if the student scores A and $Y_i = 0$, otherwise). Other explanatory variables include the Grade Point Average (GPA) of students and Personal –Computer ownership (PC = 1, if the student has a PC and, PC = 0, if the student does not have a PC).

Estimation of the Logit model gives the following results:

$$L_i = \alpha_0 + \alpha_1 \text{GP} + \alpha_2 \text{ASE} + \alpha_3 \text{PC} + u_i$$

We present the estimation procedure and interpretation of the model defined above. We can interpret the regression results using: **Logit, Odds-ratio and marginal effect.**

A. Logit Interpretation of the Estimated Logit Model

Table 15.4 below presents the logit regression result obtained using econometric software Stata14.

Table 15.4: Logistic Regression Result

Logistic regression				Number of obs	=	32
Log likelihood = -12.889633				LR chi2(3)	=	15.40
				Prob > chi2	=	0.0015
				Pseudo R2	=	0.3740
grade	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
gpa	2.826113	1.262941	2.24	0.025	.3507938	5.301432
ase	.0951577	.1415542	0.67	0.501	-.1822835	.3725988
pc	2.378688	1.064564	2.23	0.025	.29218	4.465195
_cons	-13.02135	4.931325	-2.64	0.008	-22.68657	-3.35613

We interpret the coefficients of the independent variables of the logit regression results as log of the odds-ratio. Coefficients of GPA and ASE in the Table above measure the change in the estimated logit for a unit change in the value of explanatory variable (holding other explanatory variables constant). Thus,

- ✓ The GPA coefficient of 2.826 means, with other variables held constant, that if GPA increases by a unit, on an average, the estimated logit increases by about 2.826 units, suggesting a positive relationship between the two. Holding other factors constant, as GPA increases by one point, the average logit value goes up by about 2.83, that is, the log of odds in favor of scoring “A” grade increases by about 2.83 and it is statistically significant as implied by the P-value of the Z-test.

- ✓ Holding other factors constant, as ASE increases by one point, the log of the odds-ratio increases by 0.095, but this effect is statistically not significant.
- ✓ Among students, students who have a PC compared to those who do not have PC, the log of odds in favor of scoring “A” grade increases by about 2.38 *and* it is statistically significant as implied by the P-value of the Z-test.

As you can see, all the explanatory variables have a positive effect on the logit, although statistically the effect of ASE is not significant. However, together all the explanatory variables have a significant effect on scoring “A” grade, as the LR statistic (an equivalent to F-test in linear regression) is 15.40, whose ***P-value*** is about 0.0015, which is less than 5 per cent.

Note, however, that a more meaningful interpretation is in terms of the odds-ratio, which are obtained by taking the antilog of the various coefficients of the logit model. Thus, if you take the antilog of the GPA coefficient of 2.826 you will get 16.8778 ($= \exp^{(2.826)}$). As GPA of a student rises by a unit, then the student is about 16.88 times likely to get Grade “A”, other things remaining the same. Using Stata, we can easily estimate the odds ratio for all explanatory variables as given in Table 15.5.

B. Odds-Ratio Interpretation

Table 15.5 below presents the odds-ratio regression result of the logit model.

Table 15.5: Odds Ratio Regression Result

Logistic regression		Number of obs		=	32
		LR chi2(3)		=	15.40
		Prob > chi2		=	0.0015
Log likelihood = -12.889633		Pseudo R2		=	0.3740

grade	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]	
gpa	16.87972	21.31809	2.24	0.025	1.420194	200.6239
ase	1.099832	.1556859	0.67	0.501	.8333651	1.451502
pc	10.79073	11.48743	2.23	0.025	1.339344	86.93802
_cons	2.21e-06	.0000109	-2.64	0.008	1.40e-10	.03487

The odds-ratio interpretation is as follows:

- Holding other variables constant, as GPA increases by one point, the probability of scoring Grade “A” is 16.87 times the probability of getting other grades (B, C, D, Fx, F).
- Holding other variables constant, as ASE increases by one point, the probability of scoring Grade “A” is about 1.1 times the probability of getting other grades (B, C, D, Fx, F) but it is statistically not significant.

- Holding other variables constant, students who have PC are 10.79 times more likely to score Grade “A” than other grades compared to non-PC owners, and it is statistically significant at 5% level of significance.

C. Marginal Effect Interpretation

Since the language of logit and odds ratio may be unfamiliar to some, we always compute the probability of something happening. Table 15.6 below presents the probability / marginal effect regression result of the logit model.

Table 15.6: Marginal Effects of Logit Regression Result

Marginal effects after logit							
y = Pr(grade) (predict)							
= .25282025							
variable	dy/dx	Std. Err.	z	P> z	[	95% C.I.	]
gpa	.5338589	.23704	2.25	0.024	.069273	.998445	3.11719
ase	.0179755	.02624	0.69	0.493	-.033448	.069399	21.9375
pc*	.4564984	.18105	2.52	0.012	.10164	.811357	.4375
(*) dy/dx is for discrete change of dummy variable from 0 to 1							

The marginal effect /probability interpretation of the estimated logit model given above is as follows:

- As GPA increases by one point, the probability of scoring grade “A” by an average student increase by 53%, if the other factors score average.
- As ASE increases by one point, the probability of scoring grade “A” by an average student increase by about 1.8%, assuming that the other factors score average.
- For a student who owns PC, the probability of scoring grade “A” is higher by 45.6% as compared to non-PC owners, assuming that the other factors score average.

Check Your Progress 2

1) Derive the odds-ratio for the logit model.

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2) How do you interpret the results of the logit model?

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3) Explain how the marginal effects of the logit model is interpreted.

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15.4 PROBIT MODEL

The probit model is usually used if the cumulative density function (CDF) is normal. On the other hand, the logit model uses the cumulative logistic function. In principle we can substitute the normal CDF in place of the logistic CDF.

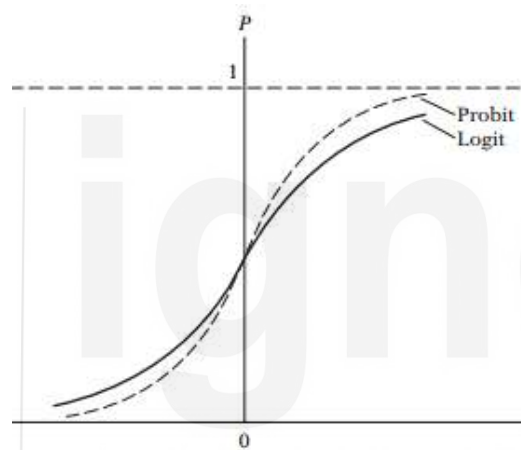


Fig. 15.3: Comparison between Logit and Probit Cumulative Distributions

Fig. 15.3 portrays the difference between the cumulative distribution function (CDF) of Logit and Probit models. As the value of the explanatory variable (X) increases, the probability of occurrence of the dependent variable increases faster (gets closer to 1) under probit model than under logit model. Similarly, as the value of X gets smaller and smaller, the probability of the dependent variable not-to-occur decreases faster (gets closer to 0) under probit than under logit. Beyond this, they are quite similar. Yet, in principle, the normal CDF can be substituted in place of the logistic CDF because of this pretty resemblance between them.

15.4.1 Specification of Probit Model

Suppose the decision of ownership of a house by a person or family depends on the utility obtained from owning the house. Mathematically,

$$I_i = \alpha_0 + \alpha_1 X_i \quad \dots (15.16)$$

where, " I_i " is *unobservable* utility index, also called **latent variable**, whose higher values are associated with higher probability of owning a house. Here, X_i is the income of i^{th} person or family. Now, assume that there is a threshold or critical level, say I_i^* , which itself is unobservable, such that if I_i **exceeds** I_i^* the family will **own a house**, otherwise it will not. If we assume further that I_i^* , like

I_i , is *normally* distributed with the same mean and variance, it is possible not only to estimate the parameters of the index given in equation (15.16) but also to get some information about the unobservable index itself.

The probability that I_i^* is **less** than or equal to I_i can be computed from the standardized normal CDF as:

$$P_i = P(Y = 1 | X) = P(I_i^* \leq I_i) = P(Z_i \leq \alpha_0 + \alpha_1 X_i) = F(\alpha_0 + \alpha_1 X_i) \quad \dots (15.17)$$

where, $P(Y = 1 | X)$ means the probability that an event occurs given the value(s) of the explanatory variables, Z_i is the standard normal variable, i.e., $Z_i \sim N(0, \sigma^2)$. Note that F is the standard normal CDF, which is:

$$\begin{aligned} F(I_i) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{I_i} e^{-\frac{z^2}{2}} dz \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\alpha_0 + \alpha_1 X_i} e^{-\frac{z^2}{2}} dz \quad \dots (15.18) \end{aligned}$$

The **area between** $-\infty$ and I_i measures the probability of **owning a house**. If we take the inverse of equation (15.18), we get information on I_i, α_0, α_1 . That is,

$$I_i = F^{-1}(I_i) = F^{-1}(P_i) = \alpha_0 + \alpha_1 X_i \quad \dots (15.19)$$

Where, F^{-1} is the inverse of the normal CDF.

In general, the probit model can be specified as follows:

$$Pr(Y_i = 1 | X_i) = Pr(I_i \geq I_i^*) = F(Z) = P_i = \int_{-\infty}^Z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z^2)} dz \quad \dots (15.20)$$

This model is called the probit model and can be estimated using maximum likelihood method (MLM). The coefficients from the probit model are difficult to interpret because they measure the change in the unobservable I_i associated with a change in one of the explanatory variables. A more useful measure is the marginal effects.

15.4.2 Estimation of Probit Model

Estimation of parameters and I_i depends on types of data we deal, meaning that whether we have grouped data or ungrouped data set.

(a) Estimation with Grouped Data (gprobit)

In the case of the logit model, we get the relative frequency, (the empirical measure of probability) of owning a house at various income level, but here in probit model, I_i can be obtained from normal CDF. Obtaining I_i makes estimating parameters relatively straightforward.

(b) Estimation with Ungrouped or Individual Data

The problem we face in Logit regression is also presents here. For this reason, we will have to use a non-linear estimating procedure based on the *method of maximum likelihood*.

15.4.3 Interpretation of Probit Model

(Example 15.3)

Let us revisit our Grade example data given in Table 15.3. It gives data on 32 students about their final grade in Mathematical Economics examination in relation to the variables GPA, ASE, and PC. Let us see what the Probit results look like. The regression results based on the method of maximum likelihood using Stata14 are given in Table 15.7.

Table 15.7: Probit Regression Result

Probit regression						Number of obs	=	32
						LR chi2(3)	=	15.55
						Prob > chi2	=	0.0014
Log likelihood = -12.818803						Pseudo R2	=	0.3775
grade	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]			
gpa	1.62581	.6938825	2.34	0.019	.2658255	2.985795		
ase	.0517289	.0838903	0.62	0.537	-.1126929	.2161508		
pc	1.426332	.5950379	2.40	0.017	.2600795	2.592585		
_cons	-7.45232	2.542472	-2.93	0.003	-12.43547	-2.469166		

Estimated coefficients do not quantify the influence of the explanatory variables on the probability that the dependent variable takes on the value one because these coefficients are parameters of the latent model. Thus, there is no standard and convincing way to interpret the coefficients from the output of a probit regression. We need to interpret the *marginal effects* of the explanatory variables, that is, how much the (conditional) probability of the outcome variable changes when you change the value of an explanatory variable, holding all other explanatory variables constant at some values. As far as the coefficients of a Probit model are concerned, what we need to do is to explain the implication of the coefficients. For instance,

- The coefficient of GPA is significantly different from zero (at 5% level) indicating relevant relationship between scoring “A” grade and student’s GPA.
- The coefficient of PC is significantly different from zero (at 5% level) indicating relevant relationship between scoring “A” grade and PC ownership.

In short, students with higher GPA and PC are more likely to score “A” compared to their counterparts. In general, “qualitatively”, the results of the Probit model are comparable with those obtained from the logit model in that GPA and PC are individually statistically significant while the variable ASE is

statistically not significant in both models. Collectively, all the coefficients are statistically significant, since the value of the **LR** statistic is 15.5458 with a p value of 0.0014.

Table 15.8: Marginal Effect of the Probit Model

Marginal effects after probit							
y = Pr(grade) (predict)							
= .26580809							
variable	dy/dx	Std. Err.	z	P> z	[	95% C.I.	]
gpa	.5333471	.23246	2.29	0.022	.077726	.988968	3.11719
ase	.0169697	.02712	0.63	0.531	-.036184	.070123	21.9375
pc*	.464426	.17028	2.73	0.006	.130682	.79817	.4375
(*) dy/dx is for discrete change of dummy variable from 0 to 1							

The marginal effect results and their interpretation in a Probit model are similar to that of logit model.

Therefore, the **marginal effect / probability** interpretation of the estimated Probit model presented above is as follows:

- As GPA increases by one unit, the probability of scoring grade “A” by an average student increases by 53%, holding other factors constant.
- As ASE increases by one point, the probability of scoring grade “A” by an average student increases by about 1.7%, holding other factors constant.
- For a student who owns PC the probability of scoring grade “A” is higher by 45.6% as compared to non-PC owners, holding other factors constant.

15.5 JOINT SIGNIFICANCE IN QUALITATIVE RESPONSE REGRESSION MODELS

As far as joint significance is concerned, as equivalent of the **F** test (in the linear regression model) there are various ways of testing multiple restrictions in Probit and Logit models.

The most commonly used test, and most easily calculated test is the **Likelihood Ratio (LR) Test**. It is used when we wish to test our exclusion restrictions, i.e. whether we should or should not exclude a set of variables. The idea is a simple one: we are maximizing is the log likelihood function, and since variables are excluded from the regression relationship, the objective function declines. The question then is whether we have a significant fall in the log likelihood function value. Like the F-test in linear regression models, in LR test the joint hypothesis to be tested is that all the explanatory variables are simultaneously irrelevant versus at least one of the explanatory variables is relevant. The likelihood ratio statistic is given as follows:

$$LR = 2 (L_{UR} - L_R) \quad \dots (15.21)$$

where, L_{UR} and L_R are the log likelihoods for the unrestricted and restricted models.

Given the null hypothesis, the LR statistic follows the chi-square (χ^2) distribution asymptotically with degrees of freedom equal to the number of explanatory variables.

15.6 GOODNESS-OF-FIT IN LOGIT AND PROBIT MODELS

A goodness-of-fit measure is a summary statistic indicating the accuracy with which the model approximates the observed data. However, the conventional measure of goodness of fit, R^2 , is not particularly meaningful in binary dependent variable models. In the case in which the dependent variable is qualitative, accuracy can be judged either in terms of the fit between the calculated probabilities and observed response frequencies or in terms of the model's ability to forecast observed responses. Measures similar to R^2 , called **pseudo- R^2** , are available. Note, however, that contrary to the linear regression model, there is no single measure for the goodness-of-fit in binary choice models and a variety of measures exists. The most common goodness of fit measure is the one that is proposed by **McFadden** (1974), which is defined as:

$$\text{Pseudo-}R^2 = \{1 - (L_{UR}/L_R)\} \quad \dots (15.22)$$

where, as before L_{UR} is the log likelihood value you obtain from the unrestricted model, and L_R is that generated by a regression (*either Probit or Logit. But note that this goodness of fit measure is also used and reported often by statistical packages for any regression that involves MLE*) with only the intercept.

- The Pseudo- R^2 measures the fit using the likelihood function and measures the improvement in the value of the log likelihood relative to having no explanatory variables (X_i). For instance, in our estimated Logit and Probit models above, we see that the Pseudo- $R^2 = 0.37$. This suggests that the log-likelihood value increases by about 37% with the introduction of the set of explanatory variables in the models.
- Similar to the R^2 of the linear regression model, it holds that $0 \leq \text{Pseudo-}R^2 \leq 1$. An increasing Pseudo- R^2 may indicate a better fit of the model, whereas there is no simple interpretation, as in the case of the classical regression model.

Check Your Progress 3

- 1) Bring out the salient features of the probit model.

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2) How do you interpret the pseudo-R-squared?

Pressure Groups

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15.7 LET US SUM UP

Here is a summary of linear probability model (LPM), logit model, and probit model:

Linear probability model (LPM): The LPM is a simple linear regression model that can be used to predict a binary outcome. The dependent variable is a dichotomous variable, such as "yes" or "no, " "success" or "failure, " or "purchased" or "not purchased." The independent variables can be continuous or categorical. The LPM assumes that the error term follows a normal distribution.

- The LPM is a simple linear regression model where the dependent variable is a binary variable.
- The coefficients in the LPM can be interpreted as the change in the mean of the dependent variable for a one-unit change in the independent variable.
- However, the LPM does not consider the fact that the dependent variable is binary, and this can lead to biased estimates of the coefficients.

Logit model: The logit model is a more sophisticated model than the LPM. It assumes that the dependent variable follows a logistic distribution. The logistic distribution is a skewed distribution that is centered at 0.5. The logit model can provide more accurate predictions than the LPM, especially when the dependent variable is close to 0 or 1.

- The logit model is a nonlinear regression model where the dependent variable is a binary variable.
- The coefficients in the logit model can be interpreted as the change in the log-odds of the dependent variable for a one-unit change in the independent variable.
- The logit model is more appropriate for modeling binary variables than the LPM because it considers the fact that the dependent variable can only take on two values.

Probit model: The probit model is another type of model that can be used to predict a binary outcome. The probit model assumes that the dependent variable follows a normal distribution. The probit model is like the logit model in terms of its predictive accuracy. However, the probit model is sometimes preferred because it is easier to interpret than the logit model.